



A.M.($\leq$)	Any	Moon P.	Gray
Image Quality		Seeing < 2.0"	
Cloudy Cover		Any	

Title of Proposal: Study of small bodies in the Solar System with stellar occultations

Telescope 1.60m Perkin-Elmer

Abstract:

A relevant characterization of small bodies in the Solar System is possible from the stellar occultation technique. Such characterization includes the determination of accurate shapes/sizes, the detection of atmospheres down to few nanobars, as well as the detection of satellites, jets, debris, rings and other structures around those objects. Continuous observations of occultation events, combined with data from different techniques, also contributes to a broad description of small bodies and to a better understanding of the history and dynamical evolution of the Solar System. Here, we propose to observe events by Quaoar (ringed TNO, needs deeper dynamical analysis of ring stability), comet Bernardinelli-Bernstein (size estimated, so far, from thermal measurements only), 2010 RO64 (TNO, never observed in an occultation), 2000 TG64 (trojan locally captured?), 2005 RM43 (large TNO from the scattered disk), Thereus (Centaur object, maybe ringed?) and Vanth, satellite of Orcus.

Category: Solar System: Small Bodies (Asteroids, Comets, Moons, Kuiper Belt, Near Earth Objects)

Time requested: 70 hours.

Minimum time accepted: 50 hours.

Observing Mode: Classical Remote (CR)

Program Type: Regular Proposal (RP)

Principal Investigator (PI): Julio Ignacio Bueno de Camargo

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Is there a thesis or dissertation that will benefit from the data?: NO

Optimal dates:

16/SEP, 01/OCT, 21/OCT, 28/NOV, 29/NOV, 11/DEC, 19/DEC

Impossible dates:

Any other

Instruments:

SPARC4 (Photometry)

Detectors	- 4 x iXon Ultra 888 CCDs
CCD Mode and Electron Multiplying (EM) Gain	- Electron Multiplier Mode: EM Gain in the interval 100-150

Scientific Justification

Small bodies in the Solar System are objects orbiting the Sun (natural satellites excluded) that do not have enough gravity to pull their shapes into a nearly spherical one. This includes (but not all) Main Belt Asteroids, Centaurs, Trans-Neptunian Objects (TNOs), comets, trojan asteroids.

TNOs and Centaurs have been our main targets at OPD, but also some trojans and comets. TNOs are further from the Sun than 30 AU (Neptune's heliocentric distance). Centaurs are objects in unstable orbits between those of Jupiter and Neptune, and they are as rich a class as TNOs alone [1]. They are thought to have, in addition, a common origin with the TNOs [2], so that Centaurs can be considered as brighter proxies of the more distant TNO population.

Their importance resides in the fact that distant small bodies are relatively unchanged relics of an evolved planetesimal disc [3-4]. Therefore, they are records of the original material that formed the solar nebula and a treasure trove to access the history and evolution of the outer Solar System. Given their faint magnitudes, there is still much to be learned from this cold and distant world.

In this study, we propose the use of the stellar occultation technique, where the flux of a star is recorded with a high temporal resolution when it is blocked by a passing small body. In this way, it is possible to characterize the body's physical properties on an individual basis that also helps to build a broader general scenario, giving a crucial view about the processes involved in the formation and evolution of the Solar System. Two striking features of stellar occultations are: possibility to translate high temporal resolution into high spatial resolution and to observe small sized objects beyond our current capability of detection (in a context of serendipitous observations).

On an individual basis, occultations deliver dimensions with kilometric accuracies [5-8]. Shapes are also accurately determined and related to density (therefore, mass), elongation, and rotation for large TNOs [8]. Accurate albedos (a measurement of the ratio between the incident and reflected light fluxes) are derived from accurate sizes and better constrain the nature of surfaces (for example, fresh ice or cometary type material, darkened by space weathering [9]). Sizes, from Kepler's laws, provide information on density for TNOs with satellites (Pluto, Eris, Haumea, etc). Densities tell about the internal structure of these objects, like the ice/rock ratio. Given the high temporal resolution in image acquisition, stellar occultations are unrivalled in providing details about the neighbourhoods of the occulting bodies (e.g., the discovery of rings around the Centaurs Chariklo and Chiron and the TNOs Haumea and Quaoar [6,19-22]). The presence of tenuous atmospheres (down to few nanobars) [5-6][10], as well as other interesting physical properties [11-15], are natural byproducts of an occultation light curve analysis.

In a general scenario, sizes of many TNOs lead to their size distribution (giving insights into the total mass in the transneptunian region) and the balance between accretion and collisional erosion [16]. Physical surface properties of TNOs tell us about the thermal and chemical processes in the outer protoplanetary disk, indicating when and where the objects formed [17] - generally, objects that formed closer to the Sun tend to display more neutral colors; objects that formed further from the Sun are redder due to, e.g., differing compositions and space weathering. Planetary migration (Neptune mainly) then placed these objects into their current places. Albedos, a basic quantity to the study of small bodies, are fundamental to constrain the concentration of ices, carbons, and organics, which can have widely differing reflectivities [18]. As mentioned above, accurate albedos can be determined from accurate diameters - a direct measurement from stellar occultations - and absolute magnitudes.

Here, we propose to observe events by the rings of the TNO (5000) Quaoar (plus main body), C/2014 UN271 Bernardinelli-Bernstein (largest Oort Cloud comet ever detected), the TNO (523640) 2010 RO64, the Trojan asteroid (47469) 2000 TG64, the large TNO (145451) 2005 RM43 from the scattered disk (SD), the Centaur (32532) Thereus and Vanth, satellite of the large TNO (90482) Orcus. New detections of Quaoar's rings help to understand their confinement and stability mechanism [20]. Comet B-B has a diameter estimated, so far, from thermal measurements only [25] and needs ephemeris improvement. (523640) 2010 RO64 is a low albedo, relatively large (diam. ~ 440 km) TNO never observed in a stellar occultation. Sizes/shapes of Trojan asteroids are important to dynamical models that test the capture of planetesimals from the Kuiper belt or local regions. Physical properties and dynamics of SD objects help to test our knowledge of the migration history of Neptune. Thereus has never been caught in an occultation event. The Orcus-Vanth is an unusual system in the Kuiper Belt region, and its physical and orbital characteristics help to distinguish possible formation scenarios [26].

Experimental Design

Sample and target selection: Targets are selected following basic criteria: visibility from OPD (do predictions indicate that the event shadow will cross the telescope site?); magnitude limit (magnitude of the occulted star and contrast between it and the occulting object); scientific interest (has the object ever been observed?; does it need more observations (to pin down size/shape, ring density/dynamics, is it targeted by a space mission, etc)?; is there any special feature - presence of ring, satellite, atmosphere, cometary activity, topography - that needs confirmation or deeper characterization?

Observing strategy: The basic point of the observational strategy is founded on the sampling of the target star flux with the highest possible temporal resolution. In other words, we want to observe the occultation event (before, during and after it) with short exposure times and ideally no dead time between consecutive images. Filters are useful to help, for instance, characterize atmospheres and ring particle size. SPARC4 has been a very useful instrument at OPD for occultations.

Observables: The observable is the flux variation of the event so that the variation of this flux, as a function of time, can be investigated. Stellar occultation translates temporal resolution into spatial resolution. Therefore, the shorter the exposure and dead times the more details about the occulted object and its neighborhoods will be contained in the light curve. Since light curves from different observatories are combined to recover the object's size and shape (and, consequently, its center), clocks should present no offsets. In this way, not only should a clock be precise but accuracy is also of the highest importance. GPS or internet protocols are frequently used to correct clocks from different sites for any offsets.

Data analysis: Achieving kilometric accuracy in determining the sizes of small solar system bodies from stellar occultations requires a meticulous approach to analyze the light curves. This involves fitting a model to the observational data that accounts for the finite exposure time, the apparent angular size of the occulted star at the distance of the occulting body and Fresnel diffraction effects (particularly important for precisely measuring features like rings).

Once the precise instants of the disappearances and reappearances of a star (ingress and egress times) are determined from the light curves of a given event, the projected shape of the object onto the sky plane can be reconstructed and, from some assumptions, the ellipsoidal figure and orientation can be determined. Here, the position of the occulted star, the object's ephemeris, the positions of the observing sites on the Earth as well as a model of the Earth's shape, are necessary.

Timing issues are frequently present in a few of the observations. These include constant time offsets, which can be corrected using a reliable site as a reference, and truncated timestamps (showing only the integer part of the second), which can be fixed - assuming that the acquisition cycle is regular - by performing a linear fit on the image-sequence $\times$ time to determine the acquisition cycle (the slope from the fit gives the cycle time). Combining data from multiple telescopes at one site can also help compensate for data gaps caused by long camera readout times.

Quantifications: SNR determination in the present context takes into consideration the magnitude drop of the event, determined when both star and small solar system body are no longer resolved, and the magnitude of the occulted star. The magnitude drop is given by the combined fluxes of the star+object and that of the object only (in the absence of an atmosphere, the total obstruction of the stellar flux is an accepted simplification). This is given by $\Delta m = -2.5 * \log(1 + 10^{(m_a - m_e) * 0.4})$, where m_a and m_e are the solar system body and star magnitudes respectively. Obviously, although it is not necessary to detect the object's flux during the observation, the flux of the occulted star and that of the sky are of fundamental importance (if the star is too faint, the magnitude drop may not be detected regardless the occurrence of the event and the magnitude of the occulting body). The value of Δm for the events proposed in this work ranges from -0.6 to -5.8. Considering that the magnitude uncertainty can be estimated from $\frac{2.5}{\ln(10)} \times \frac{1}{SNR}$, a SNR of 10, for example, is enough. The speed of the occultation shadow on the ground is also of relevance, as slow speeds accept longer exposure times without loss to spatial resolution. In practice, exposure times ranging from fractions of a second to a few seconds are used. Since occultations are serendipitous and transient events, observations are done regardless seeing, moon or the presence of clouds (limited to the safety of the telescope). Also, if necessary but at the cost of time (and spatial) resolution, images can be co-added to improve SNR. **Note:** one of the events involves a bright star (G=12.9). This is a great opportunity to acquire images at a very high rate (10 Hz or better) and profit from considerable spatial resolution.

Technical Description

Instrument configuration: High temporal resolution with adequate SNR can be obtained from a combination between the telescope aperture and a fast readout EM (100-150) camera. The latter is easily obtained from SPARC4 iXon EM detectors. For the events proposed here, the PE is the most suitable one at OPD (most of them can be considered faint - 1 star brighter than $G=15$ can profit from higher temporal resolution with good SNR).

Sky conditions: Since occultations are serendipitous and transient events, observations are done regardless seeing, moon or clouds presence (limited to the safety of the telescope). Objects may potentially hide features that only high spatial resolution can unveil and stellar occultation is the unrivaled and cheapest ground-based technique for this task. With little telescope time and simple instrumentation, moons, rings, close binary systems (where we thought to exist a single object), atmospheres and other features can be discovered at very large distances from the Sun. Therefore, image acquisition, even under poor conditions, is our preferred option.

Exposure times and SNR: The exposure time depends on factors like the magnitude of the star (frequently, the asteroid flux has little influence due to its faint magnitude and short exposure times) and the expected duration of the event. We want the best compromise between time resolution (sequence of images with short exposure times and virtually with no dead time), high SNR to accurately model the occultation light curve, and sampling inside the event. From our experience, considering clear skies and given the magnitudes of the occulted stars (see Sect. "Object List"), exposure times from < 1 s to 2-3 seconds are enough for our purposes. **Please note: We request, to the observation of the occultations, the use of the EMCCD 100-150 for 30 minutes and a readout rate of 10MHz. Exposures as short as 0.1s may also be necessary.**

Total observing time and overheads: In general, a total of 30 minutes per event is enough. In practice, we must add time to properly place the target in the FOV (in case of low density stellar fields or of a too bright star in the field) and set the instrument configuration (exposure time may be increased depending on the sky conditions, etc). In this way, 2 hours/event are enough to comfortably have everything set. Seven events involving seven objects, **listed below in order of priority** (event web pages are also provided), are proposed here. Since the minimum observing time unit offered by the OPD is one night, we request seven nights. The events by the Quaoar's rings, Comet B-B and 2010 RO64 are in **coordination with SOAR and other observatories**. A relevant backup program, complementary to this proposal, is presented later in this section.

1. Quaoar rings: 2026SEP17T00:41:00 UTC (<https://lesia.obspm.fr/lucky-star/occ.php?p=158076>)
2. Comet B-B: 2026-OCT-02T03:54:00 UTC (<https://lesia.obspm.fr/lucky-star/occ.php?p=155320>)
3. Thereus: 2026DEC12T06:30:00 UTC (<https://tinyurl.com/3utcj9vy>)
4. 2000 TG64: 2026NOV29T04:16:00 UTC (<https://tinyurl.com/ehx6zbr5>)
5. Vanth (Orcus' satellite): 2026DEC20T06:21:00 UTC
6. 2010 RO64: 2026OCT22T00:53:00 UTC (<https://tinyurl.com/mp4jcw5>)
7. 2005 RM43: 2026NOV30T05:40:00 UTC (<https://tinyurl.com/yyaz6wxz>)

Minimum observing time: The last two targets above can be considered as lower priorities. We would like to observe as many occultations as possible, taking into account the relevance of the events proposed here and the number of nights eventually granted by the TAC. However, we stress that stellar occultations offer excellent results with relatively small telescope time. We also understand that, since the OPD does not grant less than 1 night, the requested minimum requested time is 5 nights.

Backup program: Once the occultation is recorded, the remaining of the night will be used to photometric and astrometric observations. Astrometry is of utmost importance to improve the accuracy of predictions. Some of the targets of this proposal are natural targets for astrometry. The need for such astrometric task should significantly decrease as a consequence of the observations from the Vera Rubin Observatory's survey - the LSST. Photometric measurements to measure the rotational period of the occulting objects is also important. With their help, occultations observed in different epochs when the object is probed in different rotational phases, can be combined into a 3-D shape. **Please note:** the backup program and this proposal, although complementary, are independent. Occultations, as shown by various papers, are scientifically broad to justify alone a proposal in our opinion.

Figures and Tables

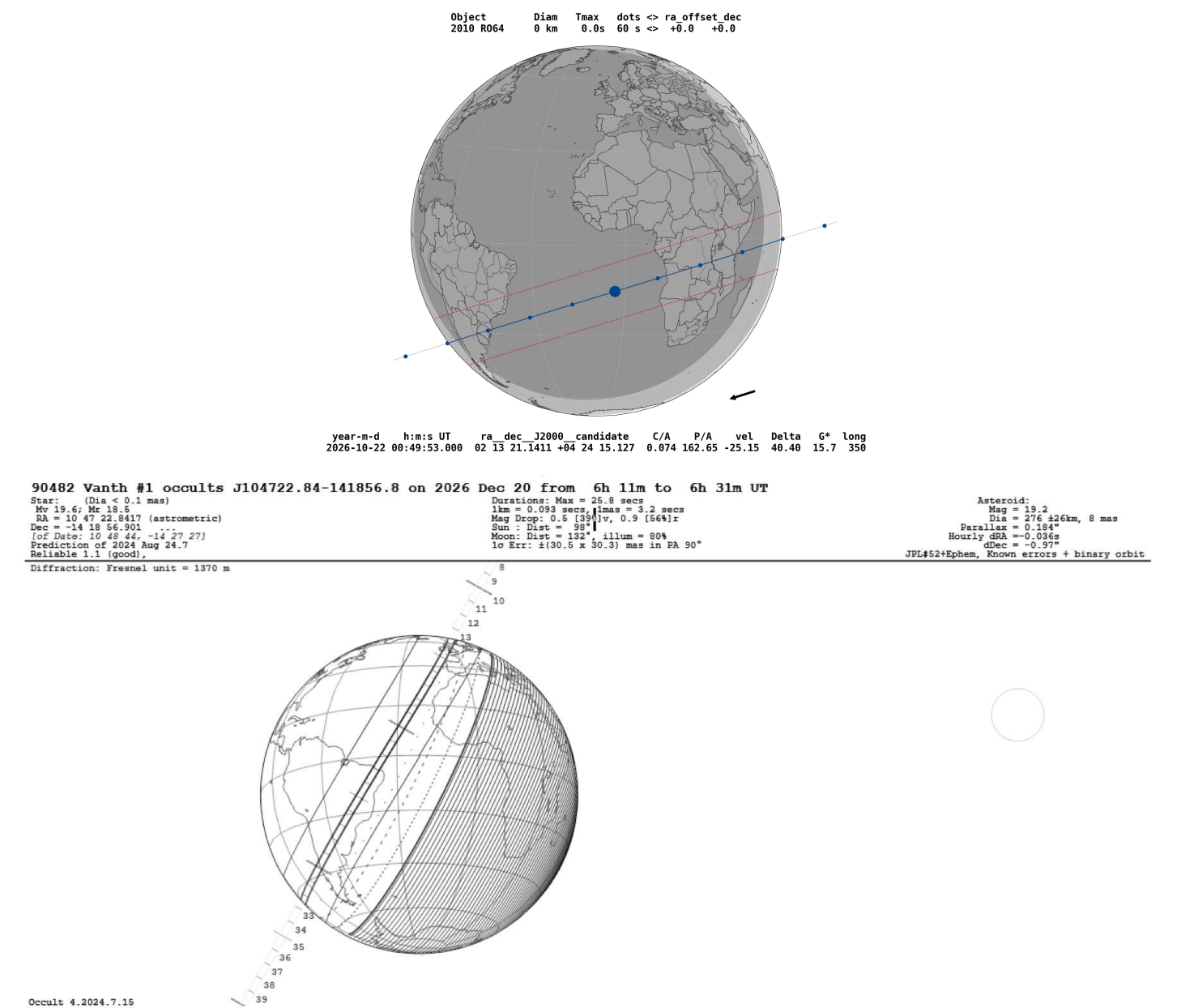


Figure 1: Two of the events in this proposal. Links to the web pages of all events are provided in Technical Description.

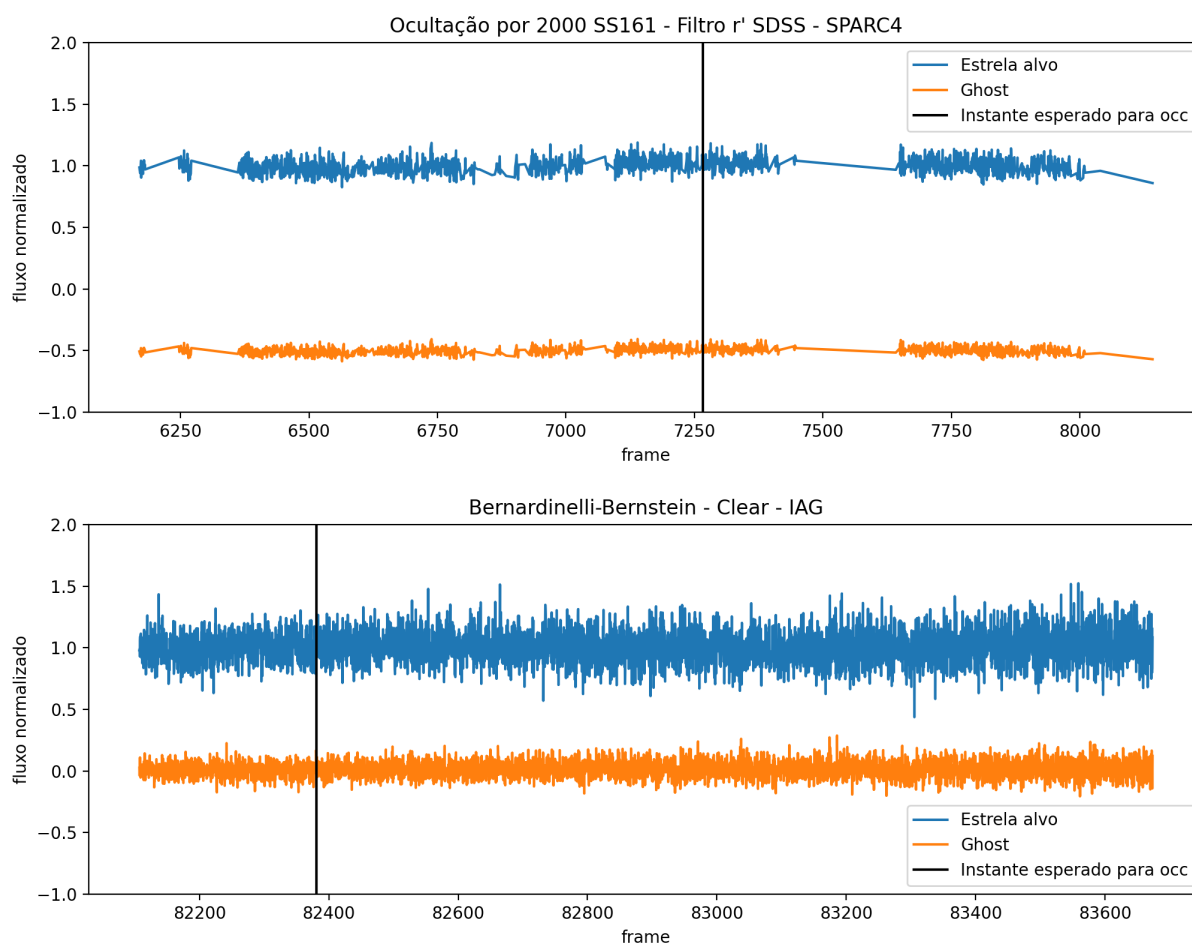


Figure 2: Light curves from the events by 2000 SS161 and comet Bernardinelli-Bernstein.

References

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Previous missions of this proposal to this telescope

Here, we report the last five proposals of this one.

- **OP2024A-006:** observations in April and July of two TNOs. One night was lost due to the weather (08/APR). The event on 19/JUL by the TNO 2002 KX14 was negative.
- **OP2024B-003:** observations in October, November and December of 2 TNOs, a comet (Bernardinelli-Bernstein) and Miranda (moon of Uranus). Three nights were lost due to bad weather, and the events by Miranda and the Neptune rings are under analysis in collaboration with a french team (Paris Observatory).
- **OP2025A-005:** observations in March, May, June and July by three TNOs, one Centaur and one Jupiter trojan. Two observations were lost due to technical problems. The events by Pholus and 2002 KX14 (TNO) were negative. The event by Dzewanna (TNO) was also probably negative but, given the presence of clouds, a careful analysis is underway before a conclusion.
- **OP2025B-002:** observations in September, November and December. The event by the large Oort Cloud comet Bernardinelli-Bernstein, was lost due to the weather. The event by 2000 SS161 (November, see Fig. 2) was negative and acquired under lots of clouds. The event by 2013 LU28 (December) was lost due to bad weather.
- **OP2026A-002:** Events in March, April, May, June, July and August by a comet (Bernardinelli-Bernstein), TNOs and a Jupiter trojan were scheduled. The event by 1996 ST1 (March - carried out the Zeiss telescope due to renovations at OPD) was not observed due to bad weather. The event by comet Bernardinelli-Bernstein (April - carried out the IAG telescope due to renovations at OPD, see Fig. 2) was negative.

Previous missions of other proposals to this telescope

- **OP2024B-002:** "*Astrometry and photometry of small bodies in the Solar System*" - Ten nights granted, 65% lost due bad weather. All data reduced. Positions were obtained and sent to <https://lesia.obspm.fr/lucky-star/predictions.php>. The results from proposals like this are important to improve the predictions of stellar occultations.
- **OP2025A-006:** "*Astrometry and photometry of small bodies in the Solar System*" - Eight nights granted, 35% lost due to bad weather. All data reduced. Positions were obtained and sent to <https://lesia.obspm.fr/lucky-star/predictions.php>. The results from proposals like this are important to improve the predictions of stellar occultations.
- **2025A-P009:** "*Mutual events of the Saturnian satellites*": Eight nights granted. We could not observe in one of those nights because the event occurred too early, after the technician's working hours (06:43 Local Time). Of the remaining nights, we failed to obtain data for only one night due to weather. The data is still under analysis.
- **2025B-P008:** "*Mutual events of the Saturnian satellites*" - Four nights were granted, of which two have already been observed, with one of them being lost due to weather. The data is still under analysis.
- **2025B-P005:** "*Multiband rotational photometry of small bodies in the Solar System*" - Four observation nights have been granted, which will be conducted in person between October 23rd and 27th.

Previous results in the field by the Principal Investigator

1. "Constraining the size, shape, and albedo of the large trans-Neptunian object (28978) Ixion with multi-chord stellar occultations", Kilic, Y., (2026), A&A, DOI: 10.1051/0004-6361/202557970 (**recent publication with observations carried out at the Pico dos Dias Observatory!**)
2. "Accurate geometric albedo, shape, and size of Hi'iaka from a stellar occultation", Fernández-Valenzuela, Estela (2025), NatCo, DOI: 10.1038/s41467-025-65749-1

3. "A high geometric albedo and small size for the Haumea cluster member (24835) 1995 SM55 determined from a stellar occultation and photometric observations", Ortiz, J.L. et al. (2025), A&A, DOI: 10.1051/0004-6361/202556498
4. "The Rings of (2060) Chiron: Evidence of an Evolving System", Pereira, C.L. et al. (2025), ApJL, DOI: 10.3847/2041-8213/ae0b6d
5. "Size and Shape of Jupiter Trojan (2207) Antenor from Stellar Occultations", Ferreira, F.S. et al. (2025), AJ, DOI: 10.3847/1538-3881/adefdc
6. "The trans-Neptunian object (119951) 2002 KX14 revealed via multiple stellar occultations", Rizos, J.L. et al. (2025), A&A, DOI: 10.1051/0004-6361/202554154
7. "Centaur 29P/Schwassmann-Wachmann 1 and its near-nucleus environment from a stellar occultation", Pereira, C.L. et al. (2025), Philosophical Transactions A, DOI: 10.1098/rsta.2024.0189
8. "A study of centaur (54598) Bienor from multiple stellar occultations and rotational light curves", Kretlow, M. et al. (2024), A&A, DOI: 10.1051/0004-6361/202450833
9. "Constraints on the evolution of the Triton atmosphere from occultations: 1989-2022", Sicardy, B. et al. (2024), A&A, DOI: 10.1051/0004-6361/202348756
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15. "The two rings of (50000) Quaoar", Pereira, C.L. et al. (2023), A&A, DOI: 10.1051/0004-6361/202346365
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17. "The multichord stellar occultation by the centaur Bienor on January 11, 2019", Fernández-Valenzuela, E. et al. (2023), A&A, DOI: 10.1051/0004-6361/202243214
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20. "Constraints on the structure and seasonal variations of Triton's atmosphere from the 5 October 2017 stellar occultation and previous observations", Marques Oliveira, J. et al. (2022), A&A, DOI: 10.1051/0004-6361/202141443
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46. "The Size, Shape, Albedo, Density, and Atmospheric Limit of Transneptunian Object (50000) Quaoar from Multi-chord Stellar Occultations", Braga-Ribas et al. (2013), ApJ, DOI: 10.1088/0004-637X/773/1/26
47. "Albedo and atmospheric constraints of dwarf planet Makemake from a stellar occultation", Ortiz, J.L. et al. (2012), Nature, DOI:10.1038/nature11597

PI and Co-Is publications relevant to this proposal

These are publications that summarize theoretical and conceptual subjects related to stellar occultations

1. "Stellar occultations by trans-Neptunian objects", Sicardy, B. et al. (2024), A&ARv, DOI: 10.1007/s00159-024-00156-x
2. "Stellar occultation by TNOs: from predictions to observations", Ortiz, J.L.(2020), The Trans-Neptunian Solar System, DOI: 10.1016/B978-0-12-816490-7.00019-9

Objects List.

Object	A.R.(h,m)	Dec.(deg)	Mag.	Band	Type	Comments
Quaoar	18:42:07.74	-15:01:32.97	17	G	Sci	Event by Quaoar and rings on 2026SET17T00:41:00 UTC
Comet Bernardinelli-Bernstein	07:35:32.37	-71:48:05.80	16	G	Sci	Occultation event by comet B-B on 2026OCT02T03:54:00 UTC
(523640) 2010 RO64	02:13:21.14	+04:24:15.13	16	G	Sci	Event by 2010 RO64 on 2026OCT22T00:53:00 UTC
(47969) 2000 TG64	05:05:44.80	+10:14:56.24	13	G	Sci	Occultation by 2000 TG64 on 2026NOV29T04:16:00 UTC
(145451) Rulina - 2005 RM43	05:50:27.25	+24:56:27.63	16	G	Sci	Occultation event by 2005 RM43 on 2026NOV30T05:40:00 UTC
(32532) Thereus	13:21:16.25	-11:09:04.16	17	G	Sci	Occultation event by Thereus on 2026DEC12T06:30:00 UTC
(90482) Vanth	10:47:22.84	-14:18:56.90	20	V	Sci	Occultation event by Vanth on 2026DEC20T06:21:00 UTC